

Experiment No. 12

Title:

Demonstration of VAPT Tools in Kali Linux using Nmap, Nikto and WPScan

Aim:

To perform Vulnerability Assessment and Penetration Testing (VAPT) using Nmap, Nikto, and WPScan in Kali Linux.

Objectives:

- To perform **network scanning using Nmap**
- To identify **open ports and services**
- To perform **web server vulnerability scanning using Nikto**
- To detect **WordPress vulnerabilities using WPScan**

Tools Required:

- Kali Linux (latest version)
- Nmap
- Nikto
- WPScan
- Internet connection
- Target system or demo website

Theory:

Vulnerability Assessment and Penetration Testing (VAPT)

VAPT is a security testing process used to identify vulnerabilities in networks, systems, and applications.

It consists of:

1. Information Gathering
2. Network Scanning

3. Vulnerability Identification
4. Security Analysis

Nmap

Nmap (Network Mapper) is an open-source network scanning tool used to discover hosts and services on a network.

Features:

- Host discovery
- Port scanning
- Service detection
- Network mapping

Nikto

Nikto is an open-source web server scanner used to detect vulnerabilities such as:

- Outdated server software
- Security misconfigurations
- Dangerous files
- Default server settings

WPScan

WPScan is a vulnerability scanner designed for **WordPress** websites.

Features:

- WordPress version detection
- Plugin vulnerability detection
- Theme vulnerability detection
- User enumeration

Procedure:

Step 1 – Start Kali Linux

Login to **Kali Linux** and open the terminal.

Step 2 – Enable and Start Web Server

To run a local web server, start **Apache HTTP Server**.

Start Apache server

```
sudo systemctl start apache2
```

Enable Apache to start automatically at boot

```
sudo systemctl enable apache2
```

Check Apache status

```
sudo systemctl status apache2
```

Expected output:

Active: active (running)

Step 3 – Verify Web Server

Open browser and type:

http://localhost

or

http://127.0.0.1

If Apache is running, the **Apache default webpage** will appear.

Step 4 – Network Scanning using Nmap

Find open ports on the local machine.

Command:

```
nmap 127.0.0.1
```

Output will show:

- Open ports
-

- Running services

Example:

PORT STATE SERVICE

80/tcp open http

Step 5 – Web Vulnerability Scanning using Nikto

Scan the web server for vulnerabilities.

Command:

```
nikto -h http://127.0.0.1
```

Nikto will display:

- Server vulnerabilities
 - Security configuration issues
 - Server information
-

Step 6 – WordPress Vulnerability Scan using WPScan

Scan a WordPress website.

Example demo site:

<https://demo.wp-api.org>

Run command:

```
wpscan --url https://demo.wp-api.org
```

Step 7 – Enumerate WordPress Users

```
wpscan --url https://demo.wp-api.org --enumerate u
```

This identifies WordPress usernames.

Step 8 – Enumerate Plugins

```
wpscan --url https://demo.wp-api.org --enumerate p
```

This detects installed plugins and vulnerabilities.

Result:

Successfully performed **Vulnerability Assessment and Penetration Testing** using:

- Nmap for network scanning
- Nikto for web server vulnerability detection
- WPScan for WordPress security analysis

Conclusion:

The experiment demonstrated how VAPT tools in Kali Linux can identify network services, web server vulnerabilities, and WordPress security issues.